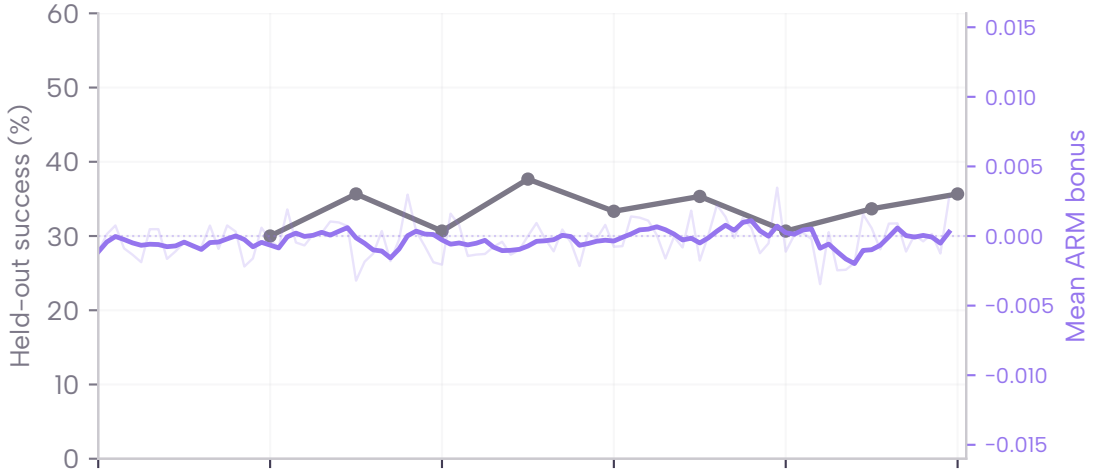
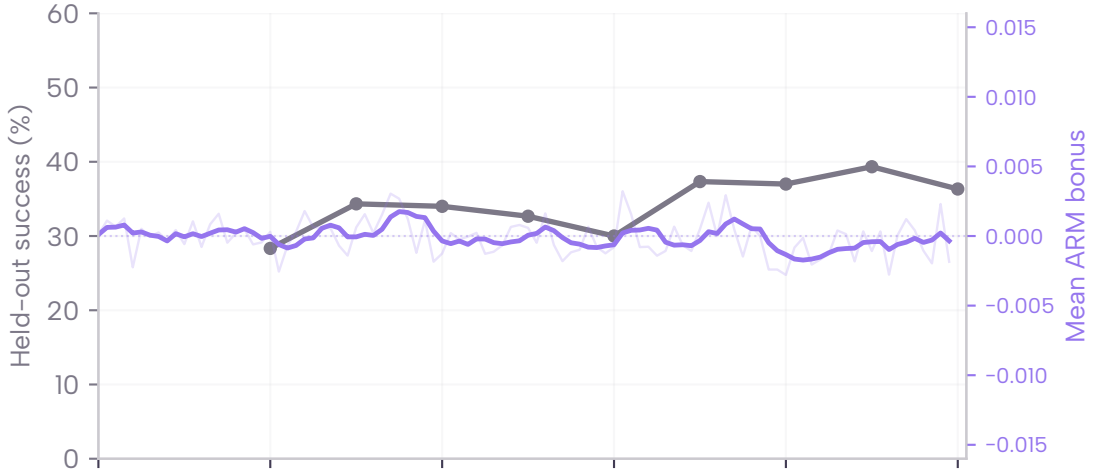
Does increasing ARM reward track task success?

Held-out full-300 success (independent judge) — Mean ARM signal (trailing 5 iterations; faint = raw)

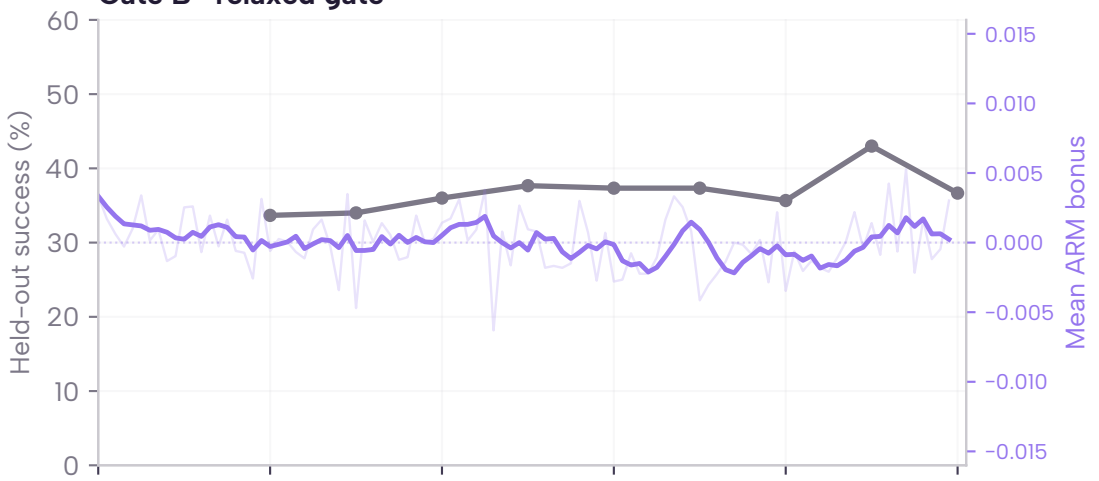
All-failure ARM



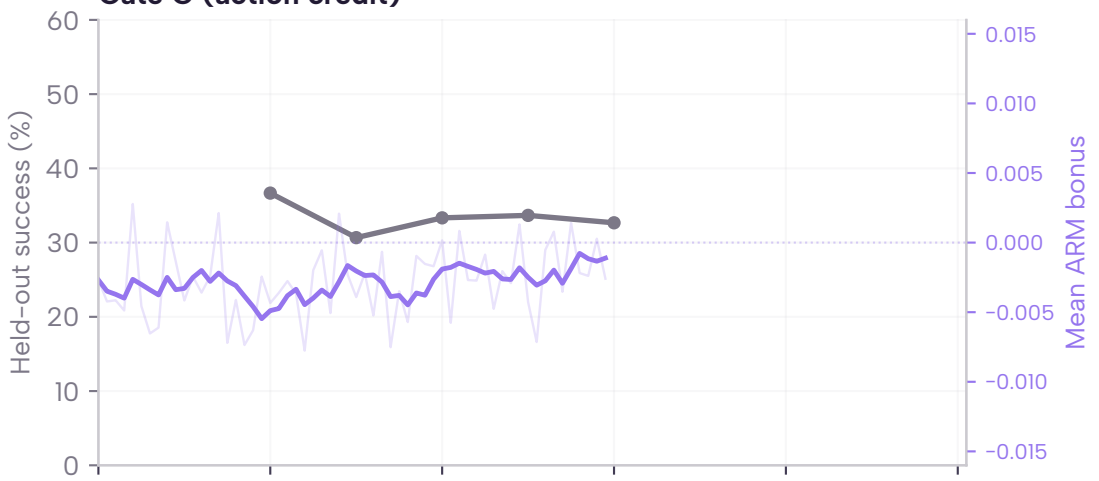
Additive ARM



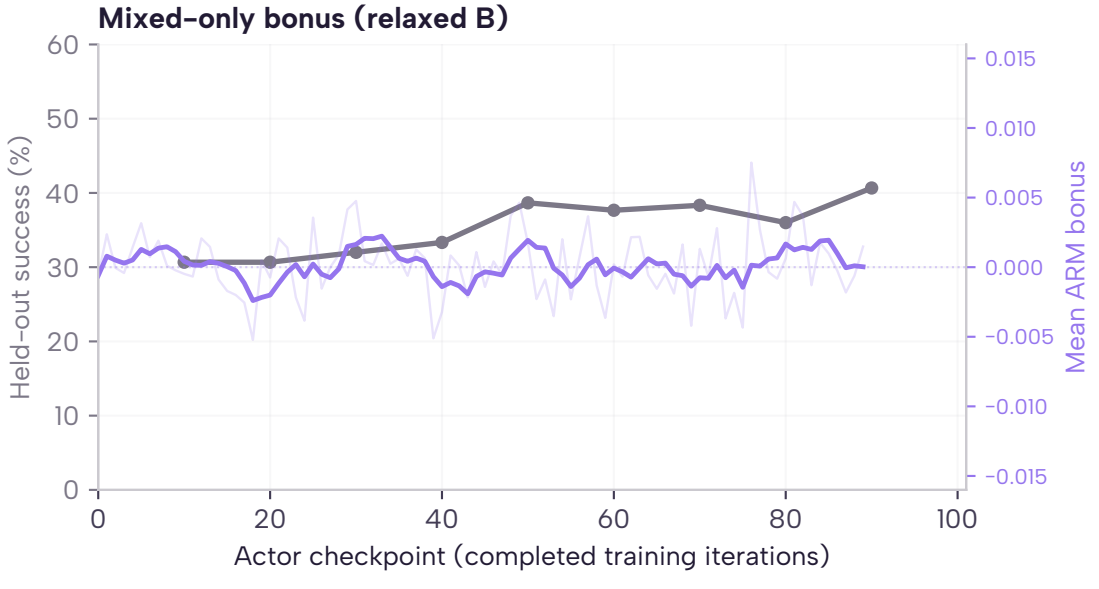
Gate B · relaxed gate



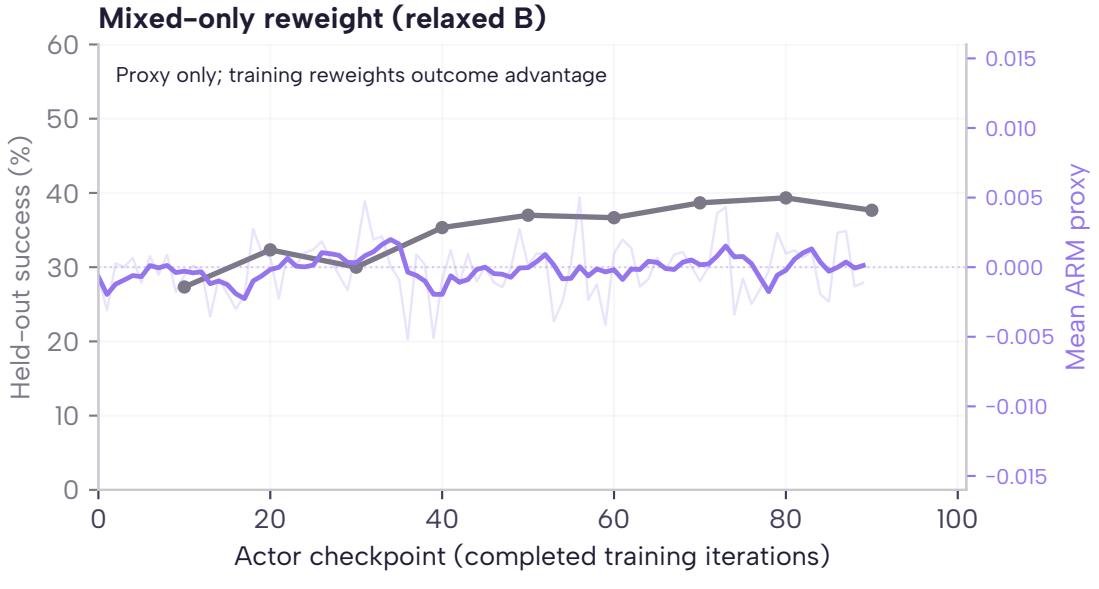
Gate C (action credit)



Mixed-only bonus (relaxed B)



Mixed-only reweight (relaxed B)



Success: local browser, GPT-4.1/action_history, actor T=0. ARM: retained main-batch turns; unlabeled turns contribute zero.
ARM is aligned to the actor BEFORE its next update. Extra failure buffers and label coverage are available in the interactive view.
Separate axes; all panels share their scales. Changing training states and live-web conditions prevent a causal reward-hacking claim.